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CO₂ Storage and Enhanced Heat/Gas Recovery in Fractured Deep High-Temperature Tight Gas Reservoirs: A Simulation Study

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Abstract

Deep reservoirs (5km-8km), with their high-temperature geological environments and sealed structural advantages, are emerging as a global research focus and potential application area for low-carbon energy transition. This paper proposes a novel approach for developing CO₂-enhanced geothermal systems (CO₂-EGS) in deep gas reservoirs. By leveraging the adsorption displacement characteristics of supercritical CO₂ to displace and enhance natural gas recovery, in situ adsorption sequestration achieves carbon geological storage. Combining the low viscosity and high diffusivity of supercritical CO₂ enhances deep geothermal energy extraction. Based on thermo-fluid dynamics and CO₂ isothermal adsorption theory, this study establishes a model for gas recovery and heat extraction through CO₂ injection and production via long horizontal wells with fracture networks. Using the Finite Element Method (FEM), the coupling of temperature-pressure equations in CO₂ horizontal wellbores and fractures with reservoir temperature fields and flow fields is solved. The impacts of wellhead injection-production parameters, fracture dimensions, and reservoir properties on geothermal energy extraction efficiency and CO₂ sequestration efficiency are analyzed. The results from the base case confirm the system's technical feasibility, demonstrating that CO₂ injection can effectively create a second gas recovery peak while establishing a stable geothermal production plateau lasting for over 15 years. A systematic sensitivity analysis reveals the key trade-offs that govern system performance. A higher CO₂ injection rate accelerates the initial recovery of gas and heat but leads to premature thermal and CO₂ breakthrough, which in turn reduces the ultimate recovery factor and shortens the system's geothermal production lifespan. A lower production bottom-hole pressure (BHP) maximizes gas recovery during the primary depletion phase but diminishes the effectiveness of the subsequent CO₂ displacement and heat extraction. Conversely, a higher BHP, while boosting the heat extraction rate, shortens the geothermal production period by increasing the system-wide pressure differential, which accelerates the advection of the cold front. Similarly, while denser fracture spacing improves the gas recovery factor by increasing the contact area, it leads to thermal short-circuiting, shortening the heat production period. Higher reservoir permeability enhances initial productivity but compromises long-term sweep efficiency and net CO₂ storage. This research provides a theoretical foundation for the design and optimization of CO₂-

enhanced geothermal systems in deep high-temperature tight gas reservoirs and offers critical insights into integrating CCUS (carbon capture, utilization, and storage) technology with geothermal energy.

Keywords: Enhanced Gas Recovery, Geothermal Energy Extraction, CO₂ Sequestration, Coupled Thermo-Fluid Dynamics, Co-optimization

Introduction

The global energy transition towards carbon neutrality necessitates the concurrent development of large-scale renewable energy sources and effective carbon reduction technologies. Deep geological formations (5--8 km), particularly high-temperature tight gas reservoirs, present a unique and compelling opportunity to address both challenges simultaneously. A promising strategy is to develop integrated systems where supercritical CO₂ is used not only to enhance geothermal energy extraction (EGS) but also to displace and recover residual natural gas (EGR) while being permanently sequestered in-situ^{[1][2][3][5][6]}. The success of this multi-objective approach, however, is critically dependent on a fundamental understanding of the coupled thermo-fluid dynamics, phase behavior, and geochemical interactions that govern the system's overall efficiency and long-term security.

The dynamics governing such an integrated system are dictated by a hierarchy of tightly coupled physical phenomena. At the fluid level, the unique thermophysical properties of supercritical CO₂---namely its low viscosity, high density, and large thermal expansion coefficient---offer superior heat extraction performance compared to water, potentially creating a "thermosiphon" effect that reduces parasitic pumping losses^{[7][8][9][10]}. Concurrently, at the fluid-rock interface, the strong adsorption affinity of CO₂ onto rock surfaces displaces adsorbed methane, enabling enhanced gas recovery and achieving secure mineralization trapping for carbon storage^{[11][12][13]}. Fundamentally, this system is characterized by a strong three-way coupling: geothermal extraction, enhanced gas recovery, and carbon sequestration are inextricably linked, and their performance must be co-optimized. Therefore, harnessing the intricate interplay between these coupled physical phenomena to achieve a predictable and synergistic outcome is the central scientific challenge.

Substantial research has been dedicated to CO₂-based EGS and EGR, though often as separate endeavors. Seminal studies on CO₂-EGS established its theoretical advantages in heat transfer efficiency over water-based systems, with subsequent work exploring the impacts of reservoir heterogeneity and injection strategies^{[7][14][15]}. Similarly, extensive research into CO₂-EGR has demonstrated its effectiveness in boosting production from unconventional resources like shale gas and coalbed methane, focusing primarily on maximizing hydrocarbon recovery through competitive adsorption mechanisms^{[16][17]}. While this body of work identified key operational and geological factors, its paradigm inherently treated heat extraction and gas recovery as independent objectives. The critical knowledge gap, therefore, is not whether CO₂ can extract heat or displace gas, but how the interplay of these processes can be managed---not as isolated goals, but as interacting components of a single, integrated system within the unique context of a deep, high-temperature, fractured gas reservoir.

Recent advances have begun to explore this integration, yet significant gaps persist in understanding the complex multi-physics environment. Numerical simulations have highlighted the potential for co-optimizing energy recovery and storage, but often rely on simplified porous media models that neglect the dominant role of fracture networks in deep tight reservoirs^{[18][19][20]}. Furthermore, while the influence of fracture geometry on fluid flow is well-studied for hydraulic fracturing, its specific role in controlling the efficiency of simultaneous heat sweeping, gas displacement, and CO₂ plume migration in a long-term geothermal operation remains poorly understood. This represents a critical barrier to the effective design and optimization of such integrated systems.

Therefore, this study aims to bridge this critical knowledge gap. We establish a comprehensive numerical model for heat and gas recovery via CO₂ injection in a deep, fractured tight gas reservoir, explicitly

accounting for long horizontal wells with complex fracture networks. Grounded in the principles of thermo-fluid dynamics and CO₂ isothermal adsorption theory, we employ the finite element method (FEM) to solve the fully coupled governing equations for the wellbore, fractures, and reservoir matrix. We then systematically analyze the influence of key controlling parameters, including wellhead injection-production conditions, wellbore and fracture geometry, and reservoir properties. The central objective is to elucidate the fundamental interplay among geothermal extraction efficiency, gas recovery, and CO₂ sequestration performance, and to identify the mechanisms that enable synergistic optimization. Ultimately, the findings are intended to provide a theoretical foundation for the design and operation of CO₂-enhanced geothermal systems, offering critical insights for integrating CCUS technology with deep geothermal energy development.

Methodology

To investigate the complex interplay of geothermal extraction, enhanced gas recovery (EGR), and CO₂ sequestration, we developed a comprehensive numerical model simulating the coupled non-isothermal multiphase flow and transport processes in a fractured reservoir. The model explicitly accounts for the distinct physical domains of the reservoir matrix, the fracture network, and the wellbore. The governing partial differential equations are solved using the Finite Element Method (FEM), providing a robust framework to handle complex geometries and strong physical couplings.

Governing Equations for Multiphase Flow and Heat Transport

The simultaneous flow of gas and supercritical CO₂, along with heat transfer in the porous reservoir matrix and fractures, is described by a set of coupled mass and energy conservation equations. The mass conservation for each component κ (where $\kappa = \text{CO}_2, \text{CH}_4$) in each phase α (where $\alpha = \text{gas, liquid}$) is given by:

$$\frac{\partial}{\partial t} \left(\phi \sum_{\alpha} S_{\alpha} \rho_{\alpha} x_{\alpha}^{\kappa} + (1 - \phi) \rho_s C_s^{\kappa} \right) + \nabla \cdot \left(\sum_{\alpha} \rho_{\alpha} x_{\alpha}^{\kappa} \mathbf{u}_{\alpha} \right) = Q^{\kappa} \quad (2.1)$$

where ϕ is the porosity, S_{α} is the saturation of phase α , ρ_{α} is the density of phase α , and x_{α}^{κ} is the mole fraction of component κ in phase α . The term $(1 - \phi) \rho_s C_s^{\kappa}$ accounts for the mass of component κ adsorbed on the rock matrix, with ρ_s being the rock density and C_s^{κ} the adsorbed concentration. $\mathbf{u}_{\alpha}$ is the Darcy velocity for phase α , and Q^{κ} is the source/sink term for component κ .

The Darcy velocity for each phase is calculated as:

$$\mathbf{u}_{\alpha} = - \frac{k k_{ra}}{\mu_{\alpha}} (\nabla p_{\alpha} - \rho_{\alpha} \mathbf{g}) \quad (2.2)$$

where k is the absolute permeability tensor, k_{ra} is the relative permeability of phase α , μ_{α} is its dynamic viscosity, p_{α} is the phase pressure, and $\mathbf{g}$ is the gravitational acceleration vector.

The overall energy conservation equation for the coupled fluid-rock system is expressed as:

$$\frac{\partial}{\partial t} \left[(1 - \phi) \rho_s C_s T + \phi \sum_{\alpha} S_{\alpha} \rho_{\alpha} U_{\alpha} \right] + \nabla \cdot \left[\sum_{\alpha} \rho_{\alpha} H_{\alpha} \mathbf{u}_{\alpha} - \mathbf{K}_{\text{eff}} \nabla T \right] = Q_H \quad (2.3)$$

where T is the temperature. The first term on the left represents the accumulation of internal energy in the rock (with density ρ_s and specific heat capacity C_s) and the fluid phases (with internal energy U_{α}). The second term describes energy transport via convection (with fluid enthalpy H_{α}) and conduction. $\mathbf{K}_{\text{eff}}$ is the effective thermal conductivity tensor of the porous medium, and Q_H is the heat source/sink term.

Flow Models

To accurately capture the role of high-conductivity pathways, the fractures and the wellbore are modeled as distinct, lower-dimensional elements that are fully coupled with the 3D reservoir matrix.

Fracture Flow: Hydraulic fractures are represented as 2D planar domains. The governing equations for mass and heat transport within the fracture plane are analogous to the reservoir equations but are integrated over the fracture aperture, w_f . The mass conservation for a component k within the fracture is:

$$\frac{\partial}{\partial t}(w_f \phi_f S_{fa} \rho_a^k x_a^k) + \nabla_T \cdot (w_f \rho_a^k x_a^k \mathbf{u}_{fa}) = q_{m \leftrightarrow f}^k + q_{w \leftrightarrow f}^k \quad (2.4)$$

where the subscript f denotes fracture properties, ∇_T is the tangential gradient operator along the fracture plane, and $\mathbf{u}_{fa}$ is the phase velocity within the fracture. The terms $q_{m \leftrightarrow f}^k$ and $q_{w \leftrightarrow f}^k$ are the mass fluxes exchanged with the matrix and the wellbore, respectively. Flow within the fracture is governed by a tangential form of Darcy's law. The fracture permeability, k_f is calculated using the cubic law for flow between parallel plates:

$$\mathbf{u}_{fa} = -\frac{k_f k_{rfa}}{\mu_a} (\nabla_T P_a - \rho_a \mathbf{g}_T) \quad (2.5)$$

$$k_f = \frac{w_f^2}{12} \quad (2.6)$$

Here, $\mathbf{g}_T$ is the component of gravity tangential to the fracture plane. A similar formulation is used for the energy balance equation within the fracture.

Domain Coupling: The system is fully coupled by enforcing the continuity of flux at the domain interfaces. The mass and heat fluxes between the matrix and fractures (and between fractures and the wellbore) are calculated based on the potential differences (e.g., pressure, temperature) between the adjacent elements. For example, the mass flux $q_{m \leftrightarrow f}$ between a matrix cell and a fracture element is governed by:

$$q_{m \leftrightarrow f} = T_{m-f} (\Phi_m - \Phi_f) \quad (2.7)$$

where T_{m-f} is a transmissibility index that depends on the geometry and permeability of the connection, and Φ_m and Φ_f are the potentials (e.g., pressures) in the matrix and fracture, respectively. This approach ensures that mass and energy are conserved across the entire model domain.

Competitive Adsorption Model

The enhanced gas recovery mechanism via CO₂ injection is critically dependent on the competitive adsorption of CO₂ and methane (CH₄) on the rock surface. This study employs the extended Langmuir isotherm to model this process:

$$C_s^k = V_{Lk} \frac{b_k p^k}{1 + \sum_{j=1}^N b_j p^j} \quad (2.8)$$

where C_s^k is the volume of component k adsorbed per unit mass of rock. V_{Lk} is the Langmuir volume constant for component k , representing the maximum adsorption capacity. b_k is the Langmuir pressure constant, indicating the strength of adsorption, and p^k is the partial pressure of component k . This formulation captures the displacement of adsorbed CH₄ as the partial pressure of CO₂ increases.

Model Setup and Simulation Scenarios

To simulate the coupled processes of CO₂ injection, gas displacement, and heat transfer, a two-dimensional (2D) numerical model was developed. This model represents a deep (~5 km), high-temperature (~170°C) tight gas reservoir, incorporating the physics of Darcy's Law, Transport of Diluted Species, and Heat Transfer in Porous Media. As depicted in Fig. 1, the model geometry features a horizontal injection-production well pair, where the injection well is intersected by a series of transverse hydraulic fractures within a naturally fractured matrix.

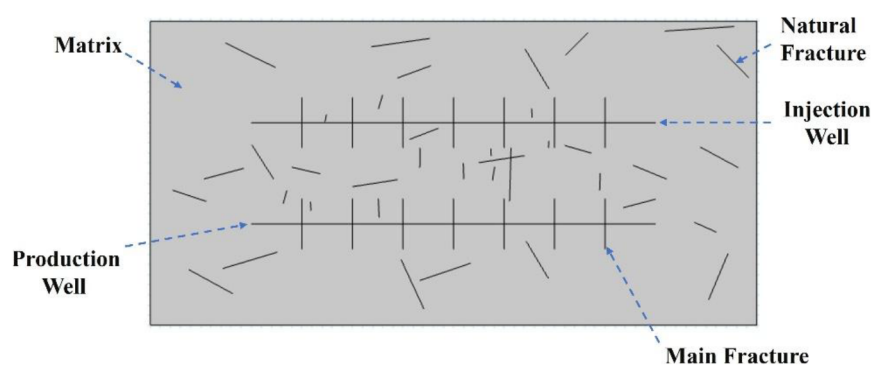


Figure 1—Schematic of the reservoir model showing the horizontal well and hydraulic fracture network. (Source: Authors)

The model is initialized with a hydrostatic pressure distribution and an initial gas saturation corresponding. The outer boundaries are treated as no-flow and no-heat-flux boundaries. The simulation encompasses two stages: an initial primary depletion phase where the production well operates at a constant bottom-hole pressure, followed by a CO₂-EGR phase. This second stage is initiated once gas production declines to a specific threshold, commencing the injection of CO₂ at a constant mass flow rate through the injection well. The key parameters for the base case simulation are listed in Table 1.

Table 1—Key parameters used in the simulation (Source: Authors)

Parameter	Value	Unit
Reservoir Properties		
Reservoir Depth	5000	m
Initial Reservoir Pressure	50	Mpa
Initial Reservoir Temperature	443.15	K
Matrix Porosity	0.052	
Matrix Permeability	6.9e-15	m ²
Rock Density (dry)	2650	kg/m ³
Rock Heat Capacity	920	J/(kgK)
Rock Thermal Conductivity	2.5	W/(mK)
Fracture Properties		
Hydraulic Fracture Permeability	1.3e-6	m ²
Hydraulic Fracture Aperture (Width)	4	mm
Hydraulic Fracture Spacing	100	m
Natural Fracture Permeability	5e-11	m ²
Natural Fracture Aperture (Width)	0.4	mm
Model Size	1200*600	m

To comprehensively assess the system's performance, a systematic sensitivity analysis was conducted. This analysis investigated the influence of: (i) injection/production conditions; (ii) fracture geometry and (iii) reservoir properties. The impacts of these variations on heat extraction efficiency, gas recovery factor, and CO₂ sequestration volume were systematically evaluated.

Results

Base Case Analysis: Unveiling the Mechanisms of Synergistic Gas and Heat Recovery

This section presents a detailed analysis of the base case simulation, whose parameters are listed in Table 1. The objective is to illustrate the fundamental thermo-fluid dynamics throughout the two primary operational stages: (1) primary depletion, and (2) CO₂ injection for enhanced gas recovery (EGR) and geothermal heat extraction.

The simulation begins with a primary depletion phase, where the production well operates at a constant bottom-hole pressure. As illustrated in Fig. 2, the production dynamics are characteristic of a fractured tight reservoir: the CH₄ production rate ramps up to a peak of approximately 4.86 kg/s around day 80, followed by a steep, hyperbolic decline. However, after approximately 2.5 years (900 days), the production rate falls to 0.1 kg/s. This point represents a critical threshold of diminishing returns, where the primary drive mechanism is significantly weakened, and continuing with depletion would lead to a prolonged period of low-efficiency "tail production." Delaying intervention further would be economically suboptimal. Therefore, this study identifies day 900 as the strategically optimal moment to transition from primary recovery. At this juncture, CO₂ injection is initiated to proactively enhance gas and heat recovery, capitalizing on the remaining reserves before they become uneconomical to produce, and concurrently beginning the vital process of in-situ carbon sequestration.

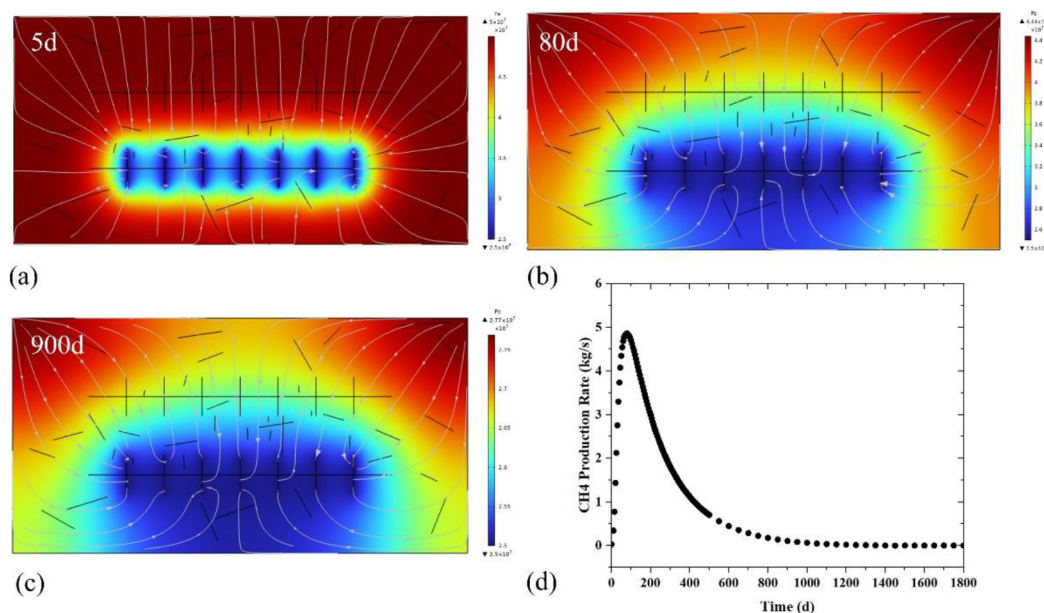


Figure 2—Production dynamics and reservoir pressure evolution during the primary depletion phase. (a-c) Reservoir pressure distribution maps at 5, 80 (peak production), and 900 days, respectively. (d) The corresponding CH₄ production rate curve, showing the initial ramp-up, peak, and subsequent decline. (Source: Authors)

Following the 900-day primary depletion phase, supercritical CO₂ is injected into the designated injection well at a constant mass flow rate of 770 t/d, marking the transition to the enhanced recovery stage. As depicted in Fig. 4, the previously declining CH₄ production rate experiences a sharp resurgence, forming a distinct secondary production peak that surpasses the prior economic threshold. This significant production enhancement is attributed to the combined action of three synergistic mechanisms: First, the injected CO₂ rapidly pressurizes the reservoir, particularly along the high-permeability fracture network. This process re-energizes the system, providing crucial pressure support to drive the remaining free-phase CH₄ towards the production well. Second, as the CO₂ plume expands from the injector, it acts as an effective displacement agent, physically sweeping the mobile CH₄ ahead of it in a piston-like manner. Finally, and critically for

tight gas reservoirs, the mechanism of competitive adsorption comes into play. Due to its higher adsorption affinity for the organic-rich matrix surface, CO_2 molecules displace previously adsorbed CH_4 , liberating additional gas that would otherwise be trapped and unrecoverable by pressure depletion alone. These three mechanisms work in concert to maximize the recovery of the remaining gas-in-place.

Fig. 3 details the spatio-temporal evolution of physical fields, including pressure, temperature, and gas composition, within the reservoir following CO_2 injection. The pressure field (Fig. 3a) shows that the injection process establishes a continuously propagating high-pressure front, creating a stable pressure gradient from the injection to the production well; this effectively reverses the reservoir-wide pressure depletion trend from the primary recovery phase and provides a sustained driving force for enhanced recovery. Regarding the temperature field (Fig. 3b), the injection of cooler CO_2 (20°C) into the high-temperature reservoir (170°C) forms a distinct "cold plume," whose thermodynamic front's gradual propagation towards the production well along the fracture network is predominantly governed by advective heat transfer—the core mechanism of geothermal extraction. The gas composition field (Fig. 3c) clearly visualizes the displacement process, where the advancing CO_2 plume preferentially sweeps along high-conductivity fractures, forming a mobile, enriched CH_4 "gas bank" ahead of it. Critically, this advective process is coupled with mass transfer into the surrounding rock matrix, as CO_2 diffuses from fractures into the low-permeability matrix over time, which not only ensures its effective sequestration but also displaces additional CH_4 from the matrix pores, thereby fully demonstrating the powerful synergy between enhanced gas recovery and carbon sequestration.

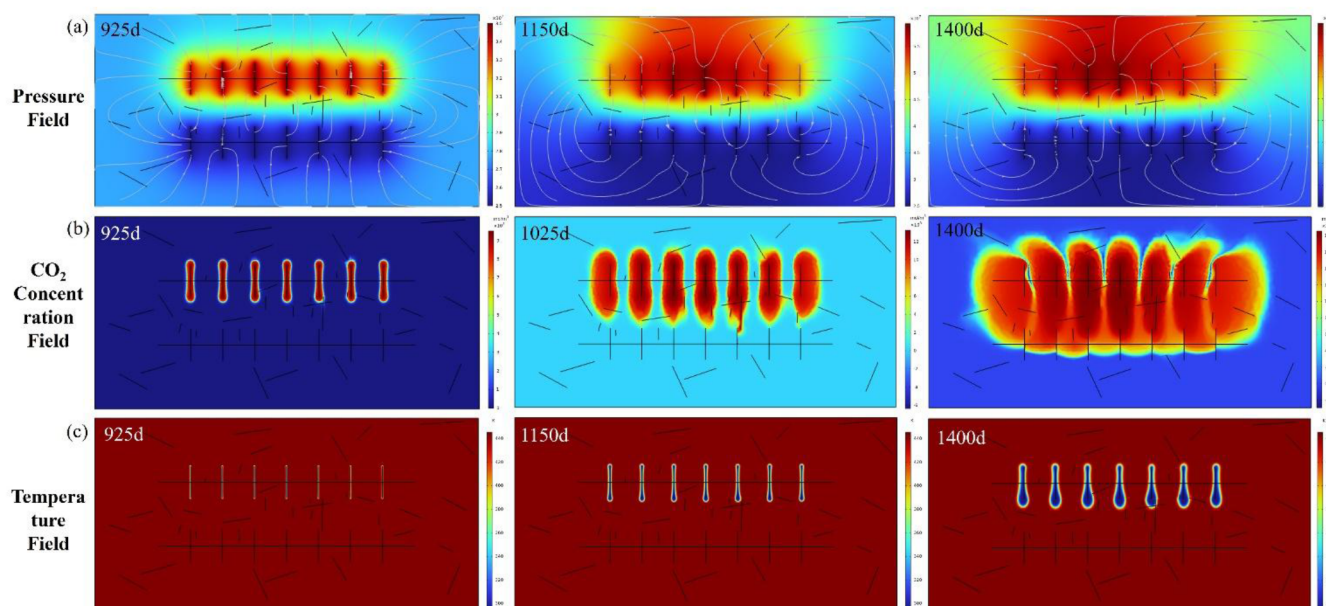


Figure 3—Spatio-temporal evolution of reservoir physical fields during CO_2 injection. (Source: Authors)

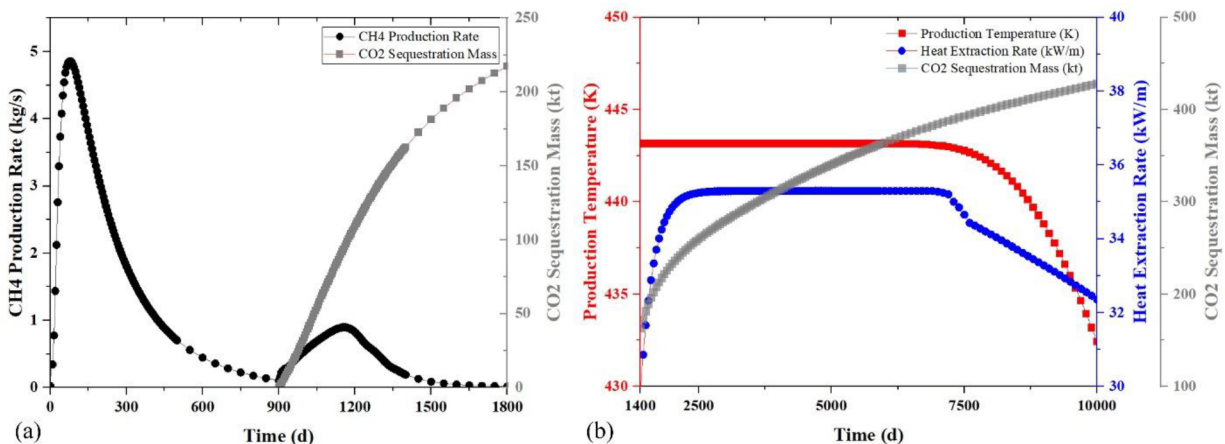


Figure 4—System performance indicators during the CO₂-enhanced recovery and geothermal extraction stage. (a) CH₄ production rate and cumulative mass of sequestered CO₂ over time (0-1800d). (b) Corresponding production temperature and heat extraction rate at the production well (1400-10000d). (Source: Authors)

Fig. 4 quantitatively illustrates the overall system performance during the CO₂ injection stage. As shown in Fig. 4a, CH₄ production experiences a rapid resurgence after injection commences on day 900, reaching a second peak of 0.89 kg/s before declining to nearly zero around day 1400 due to CO₂ breakthrough at the production well, demonstrating a significant enhancement in gas recovery. Concurrently, the CO₂ sequestration process exhibits two distinct phases: an initial rapid storage period (days 900-1800), where 217 kt of CO₂ is sequestered within 900 days, followed by a period of slower storage (days 1800-10000), where an additional 210 kt is sequestered over the next 8200 days. This reduced rate is attributed to the slowing expansion of the CO₂ plume's swept volume and the partial production of injected CO₂ for heat extraction. In terms of geothermal extraction (Fig. 4b), the production temperature remains high, near the initial reservoir temperature of 443 K, for an extended period until approximately day 7000. Thereafter, as the cold CO₂ front reaches the production well, both the produced fluid temperature and the heat extraction power decline sharply. Although the system's heat extraction power is 30-40 kW/m lower than that of a pure CO₂-EGS system with a more complex fracture network, it provides a stable heat production plateau lasting approximately 15 years. This result strongly confirms the technical feasibility of synergistically combining CO₂-enhanced gas recovery, long-term geological sequestration, and geothermal energy utilization in deep tight gas reservoirs.

Impact of Injection and Production Conditions

Having established the fundamental feasibility of the integrated process in the base case (Section 3.1), where a clear synergy between enhanced gas recovery, geothermal extraction, and carbon sequestration was demonstrated, the subsequent critical step is to understand how this performance can be actively controlled and optimized. In any field application, the primary levers for managing this complex interplay are the operational conditions at the injection and production wells. Therefore, this section presents a systematic sensitivity analysis investigating the influence of key parameters: the CO₂ injection rate and the production well's bottom-hole pressure (BHP).

Effect of Injection Rate. The CO₂ injection rate is a primary operational lever that governs the system's dynamic response and the interplay between recovery rates and long-term efficiency. To investigate this, three injection rates were simulated: 300 t/d (low), 770 t/d (base case), and 1200 t/d (high).

The results, presented in Fig. 5, reveal that a higher injection rate significantly accelerates the system's initial performance but at the cost of long-term productivity and efficiency. As shown in Fig. 5a, increasing the injection rate from 300 t/d to 1200 t/d leads to a more pronounced and earlier secondary CH₄ production

peak. This is a direct consequence of the intensified pressure support and faster displacement front propagation, which more rapidly pushes the mobile CH_4 bank towards the production well. Similarly, the initial heat extraction rate is highest for the 1200 t/d case (Fig. 5d), as the greater mass flux through the production well enhances convective heat transport before thermal breakthrough. However, this accelerated recovery introduces a critical trade-off related to premature breakthrough. The high-velocity CO_2 front in the 1200 t/d case channels through the high-permeability fracture network, leading to a much earlier CO_2 breakthrough. This effect is twofold. First, it abruptly curtails the CH_4 production phase, potentially trapping significant volumes of gas in the matrix due to fluid bypassing and reducing the ultimate gas recovery. Second, it causes an early thermal breakthrough, as clearly depicted by the rapid decline in production temperature for the 1200 t/d case around day 5000 (Fig. 5c). Consequently, the stable heat production plateau is drastically shortened, diminishing the system's geothermal lifespan.

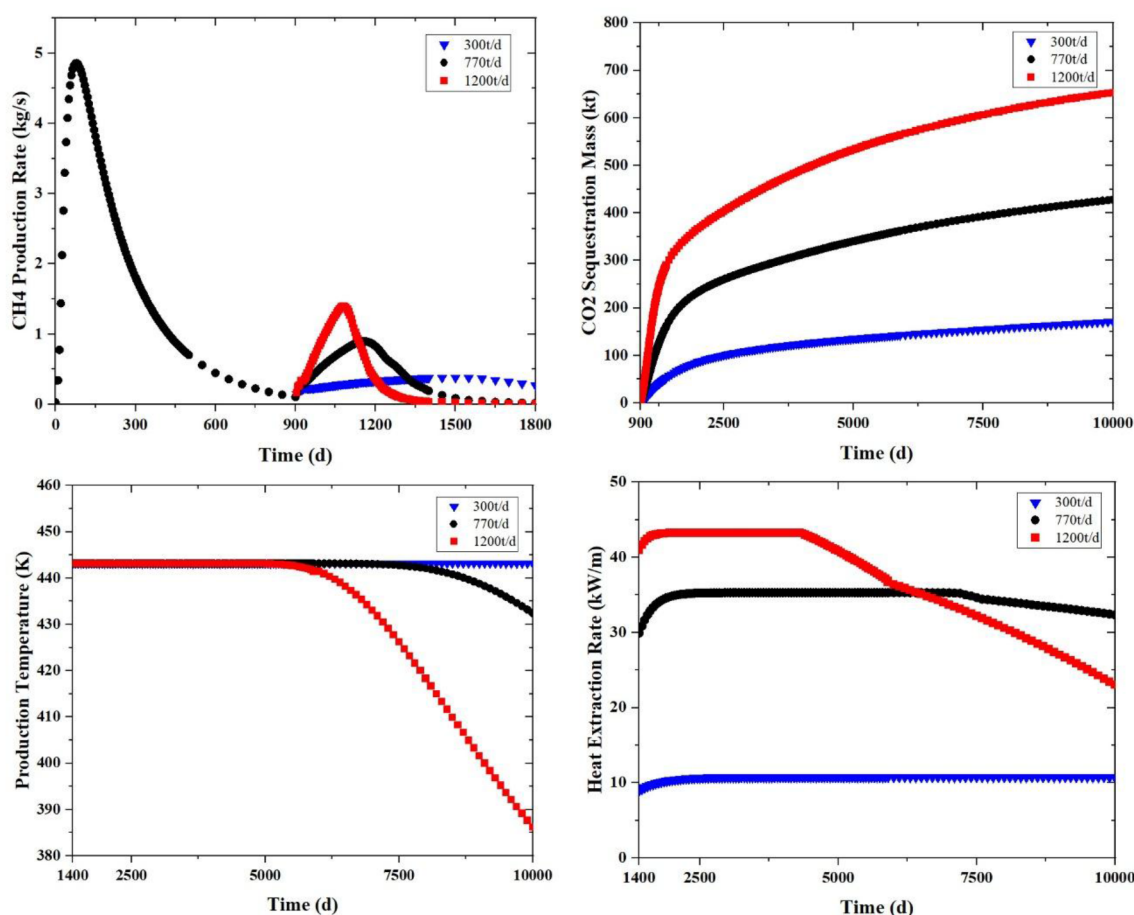


Figure 5—Impact of CO_2 injection rate on key performance indicators: (a) CH_4 production rate, (b) cumulative mass of sequestered CO_2 , (c) production temperature, and (d) heat extraction rate. (Source: Authors)

In essence, these results highlight a critical operational trade-off: while a high injection rate accelerates initial gas and heat recovery, it does so at the expense of sweep efficiency and the system's geothermal lifespan. The premature breakthrough of CO_2 not only curtails the EGR process, potentially reducing ultimate gas recovery, but also shortens the duration of stable heat production. This underscores that an optimal injection strategy must balance the immediate gains in production rate against the long-term objectives of maximizing ultimate recovery and sustaining the geothermal heat plateau.

Effect of BHP. The bottom-hole pressure (BHP) of the production well is a crucial operational parameter that dictates the pressure drawdown, governing the interplay between the rate of resource extraction and

the overall sweep efficiency. To investigate this, three BHP levels were simulated: 15 MPa (low), 25 MPa (base case), and 35 MPa (high).

The simulation results, illustrated in Fig. 6, reveal a critical trade-off between the rate of recovery and the long-term sustainability of the process. Specifically, Fig. 6a shows that a lower BHP (15 MPa) accelerates primary recovery, yielding a higher initial CH_4 peak by establishing a greater pressure drawdown. The consequence of this aggressive initial depletion becomes apparent during the subsequent CO_2 -EGR phase: since more CH_4 has been produced upfront, the potential for CO_2 -driven recovery is subsequently reduced, leading to a less effective displacement process. A higher BHP (35 MPa) leads to a significantly earlier thermal breakthrough, with the temperature beginning to decline around day 6000 (Fig. 6c). This phenomenon can be directly explained by the pressure field dynamics. Our simulation shows that under a constant injection rate, the stabilized pressure differential between the injector and producer is actually larger in the high BHP case ($\Delta P \approx 47.7$ MPa) than in the low BHP case ($\Delta P \approx 43.2$ MPa). This larger driving force (ΔP) accelerates the advective transport of the cold CO_2 front within the fractures, causing it to reach the production well faster. Concurrently, the 35 MPa case also demonstrates superior CO_2 sequestration performance (Fig. 6b).

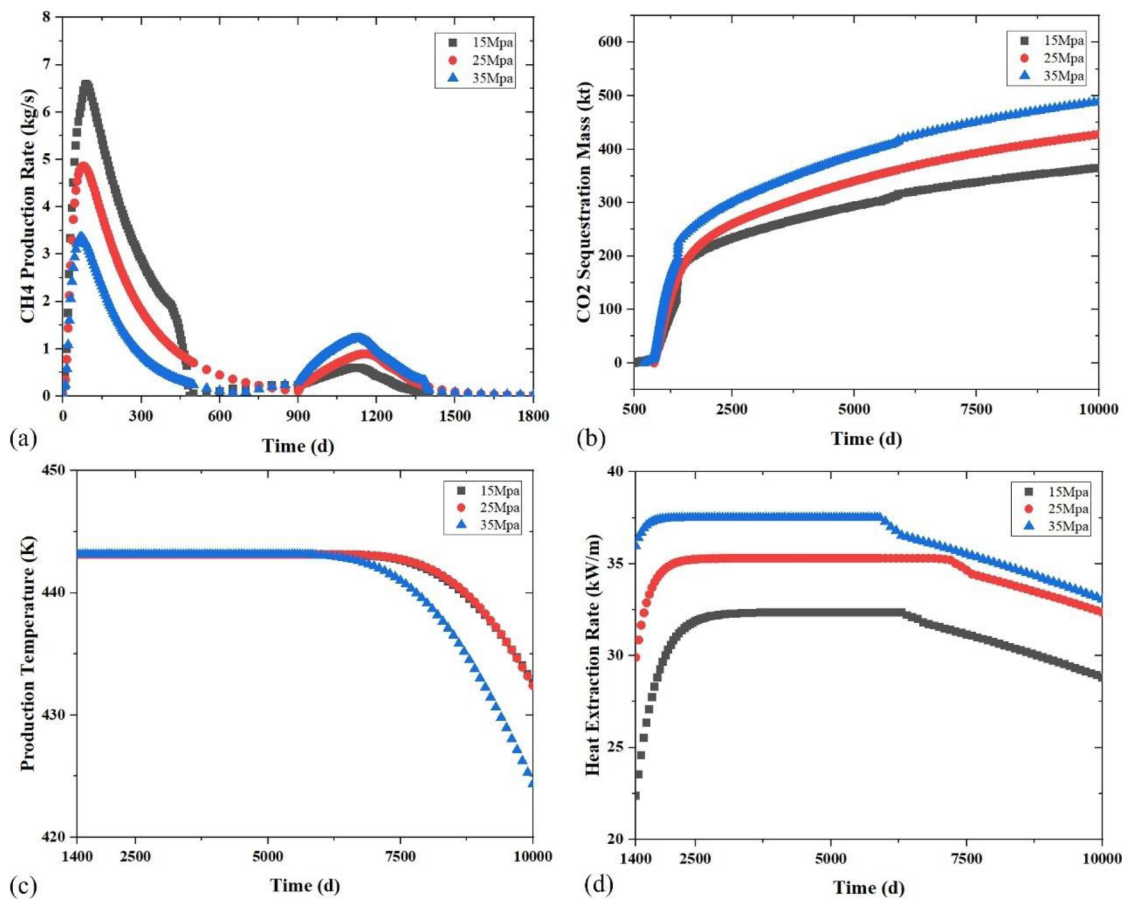


Figure 6—Impact of BHP on key performance indicators: (a) CH_4 production rate, (b) cumulative mass of sequestered CO_2 , (c) production temperature, and (d) heat extraction rate. (Source: Authors)

In summary, the production BHP presents a complex operational trade-off between initial recovery and the management of the geothermal resource. A low BHP strategy maximizes immediate gas production during primary depletion, but this comes at the cost of a less effective subsequent CO_2 -EGR phase and a lower rate of heat extraction, despite a longer thermal production plateau. Conversely, a high BHP strategy leads to a paradoxical outcome: a higher heat extraction rate but a significantly shorter geothermal

lifespan. This is driven by the larger system-wide pressure differential required at high backpressure, which accelerates the advective transport of the cold front. Therefore, the optimal BHP must be selected based on a strategic evaluation of whether to prioritize rapid upfront gas recovery or to optimize the rate and duration of geothermal energy extraction.

Impact of Fracture spacing

Hydraulic fracture spacing is a primary design parameter in reservoir stimulation. We analyzed three scenarios: 150 m (sparse), 100 m (base case), and 50 m (dense). The impact on CH₄ recovery and cumulative heat extraction is shown in Fig. 7.

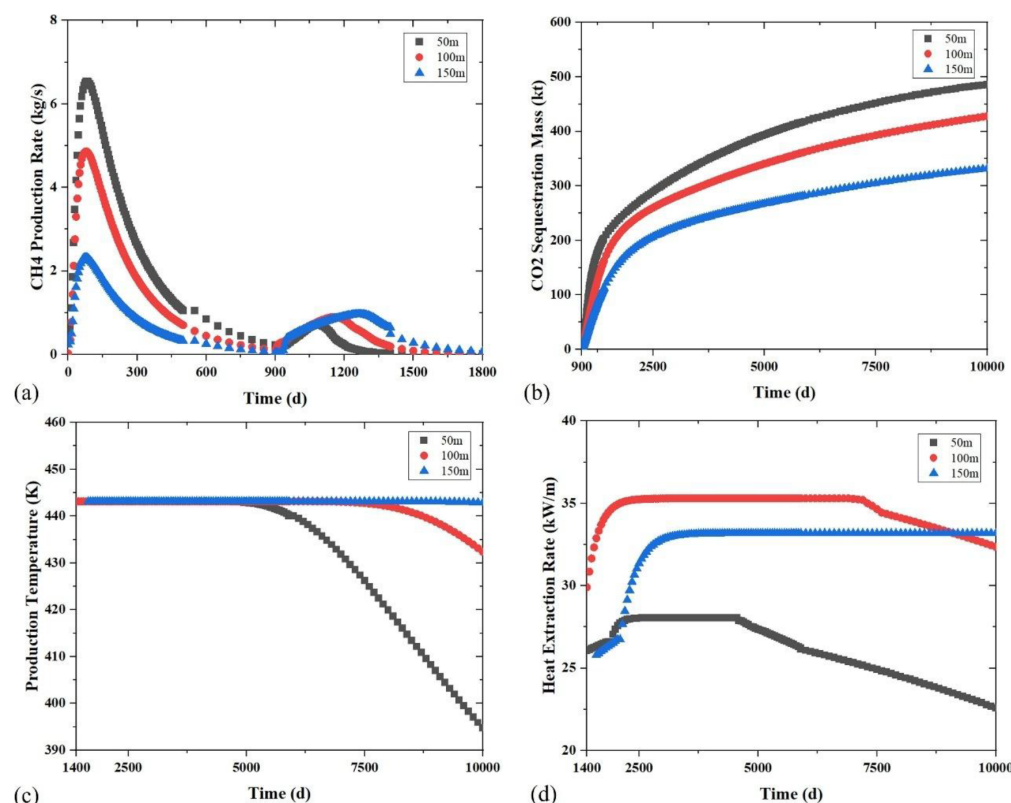


Figure 7—Impact of hydraulic fracture spacing on key performance indicators: (a) CH₄ production rate, (b) cumulative mass of sequestered CO₂, (c) production temperature, and (d) heat extraction rate. (Source: Authors) h3@Impact of Reservoir Properties

The simulation results clearly indicate that denser fracture spacing accelerates resource recovery, but at the cost of long-term geothermal sustainability. As shown in Fig. 7a, decreasing the fracture spacing from 150 m to 50 m leads to a higher and earlier primary CH₄ production peak. This is because a denser fracture network provides a larger contact area, enabling more efficient and rapid collection of CH₄. It also results in a slightly higher net CO₂ sequestration (Fig. 7b). However, the dense fracture network also creates a more direct and efficient channel for the injected cold fluid to travel from the injector to the producer. This minimizes the volumetric heat exchange with the surrounding rock matrix and consequently exacerbates thermal breakthrough. Fig. 7c demonstrates that the 50 m spacing case experiences a dramatically premature thermal breakthrough, with production temperatures beginning to decline before day 5,000. In contrast, the 150 m spacing case maintains a stable high temperature for over 10,000 days. Consequently, the stable heat production period is significantly shortened in the dense fracture scenario (Fig. 7d).

In summary, fracture geometry presents a fundamental design trade-off. While a dense fracture network (e.g., 50 m spacing) is highly advantageous for maximizing the CH₄ recovery rate by increasing the

fluid-rock contact area, excessively dense fracturing can lead to premature thermal short-circuiting, which shortens the system's geothermal production lifespan. Therefore, a balanced approach that weighs these competing outcomes is essential for optimal design.

Impact of Reservoir Properties

Reservoir permeability is a fundamental geological property that dictates the fluid flow dynamics and thus governs the interplay between resource recovery efficiency and long-term storage security. To investigate its impact, three matrix permeability values were simulated: $6.9\text{e-}16\text{ m}^2$ (low), $6.9\text{e-}15\text{ m}^2$ (base case), and $6.9\text{e-}14\text{ m}^2$ (high).

As shown in Fig. 8a, the case with the highest permeability (69 mD) exhibits a higher CH₄ production peak, suggesting that the injected CO₂ effectively displaces gas from the matrix. However, this is accompanied by a shorter production plateau and ultimately results in lower utilization of the reservoir's reserves. Conversely, the lowest permeability case (0.69 mD) yields a lower gas rate but maintains a comparatively longer stable production period. This leads to a situation, as suggested by Fig. 8d, where the heat production phase begins around day 1200 for the high-permeability case, while it is delayed until after day 3000 for the low-permeability case. According to Fig. 8b, higher permeability allows more CO₂ to be produced back through the wellbore instead of being stored, leading to poorer net sequestration. Concurrently, as seen in Figs. 8c and 8d, the higher fluid velocities in the high-permeability case result in a greater heat extraction rate but also cause an earlier thermal breakthrough.

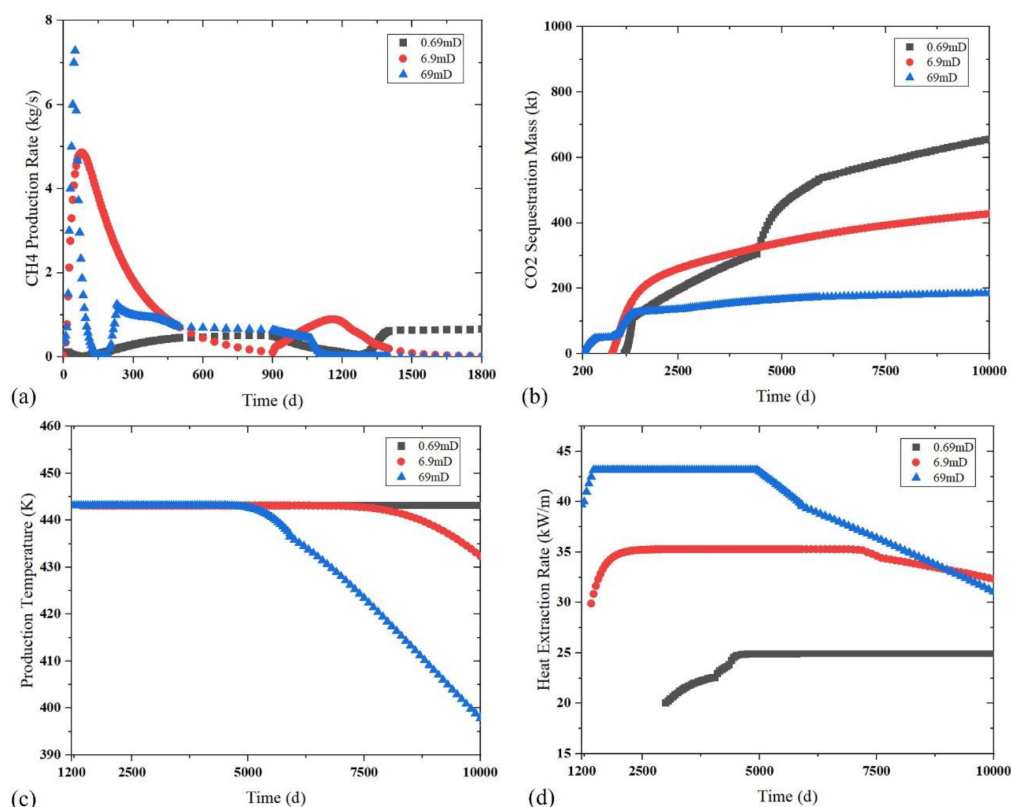


Figure 8—Impact of reservoir permeability on key performance indicators: (a) CH₄ production rate, (b) cumulative mass of sequestered CO₂, (c) production temperature, and (d) heat extraction rate. (Source: Authors)

In summary, reservoir permeability presents a key trade-off governing project performance. High permeability is highly advantageous for maximizing short-term productivity, but it compromises long-term resource recovery and CO₂ sequestration security. Conversely, while low permeability ensures long-term production stability and effective sequestration, it typically comes at the cost of lower overall productivity.

Conclusions and Recommendations

This simulation study systematically investigated the coupled multi-physics of integrating CO₂-enhanced gas recovery, geothermal energy extraction, and geological carbon sequestration in a deep, fractured high-temperature tight gas reservoir. Based on a comprehensive numerical model that accounts for the interplay between the wellbore, fractures, and reservoir matrix, the following key conclusions are drawn:

1. **Synergistic Feasibility Confirmed:** The core concept of a three-in-one system is technically feasible. The injection of supercritical CO₂ following primary depletion successfully initiates a secondary gas recovery phase, driven by reservoir pressurization, fluid displacement, and competitive adsorption. Simultaneously, it establishes a long-term, stable geothermal production plateau (lasting over 15 years in the base case), confirming the powerful synergy between enhanced resource recovery and energy production.
2. **Operational Conditions Present a Rate-vs-Efficiency Trade-off:** Both CO₂ injection rate and production well BHP are critical operational levers that present a trade-off between short-term productivity and long-term efficiency. Higher injection rates accelerate initial gas and heat recovery but lead to premature CO₂ and thermal breakthrough, significantly reducing the ultimate gas recovery and shortening the system's geothermal lifespan. Similarly, a low BHP maximizes primary gas recovery but diminishes the effectiveness of the subsequent CO₂-EGR and heat extraction. A high BHP, conversely, improves the heat extraction rate but paradoxically accelerates thermal breakthrough due to a larger system pressure differential.
3. **Hydraulic fracture spacing is a key design parameter that reveals a fundamental conflict between maximizing hydrocarbon recovery and sustaining geothermal production.** Denser fracture networks significantly improve methane recovery by increasing the reservoir contact area. However, excessively dense fracturing leads to premature thermal short-circuiting, thereby shortening the system's geothermal production lifespan. Therefore, a balanced approach that weighs these competing outcomes is essential for optimal design.
4. **Higher permeability promotes greater fluid mobility, resulting in a higher heat production rate, but this comes at the cost of earlier thermal breakthrough and reduced net CO₂ sequestration, as more CO₂ is produced back instead of being permanently stored in the matrix.**

Based on the above conclusions, future research and practical applications should focus on the following aspects. First, multi-objective optimization methods should be employed to strike a balance between enhancing natural gas recovery, ensuring geothermal sustainability, and achieving long-term CO₂ sequestration. Second, adaptive operational strategies should be explored, such as dynamically adjusting CO₂ injection rates or cycling production pressures, to prolong geothermal production lifespans and improve CO₂ sequestration efficiency. Lastly, techno-economic assessments and environmental impact analyses should be integrated to provide practical guidance and policy support for the large-scale implementation of this three-in-one system.

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Nomenclature

- ϕ = Porosity, -
 S_{α} = Saturation of phase α , -
 ρ_{α} = Density of phase α , kg/m³

- x_{α}^{κ} = Mole fraction of component κ in phase α , -
 ρ_s = Rock density, kg/m³
 C_s^{κ} = Adsorbed concentration of component κ , mol/kg
 u_{α} = Darcy velocity for phase α , m/s
 Q_{κ} = Source/sink term for component κ , kg/(m³·s)
 k = Absolute permeability tensor, m²
 k_{ra} = Relative permeability of phase α , -
 μ_{α} = Dynamic viscosity of phase α , Pa·s
 p_{α} = Phase pressure, Pa
 g = Gravitational acceleration vector, m/s²
 C_s = Specific heat capacity of rock, J/(kgK)
 T = Temperature, K
 U_{α} = Internal energy of phase α , J/kg
 H_{α} = Enthalpy of phase α , J/kg
 K_{eff} = Effective thermal conductivity tensor, W/(m·K)
 Q_H = Heat source/sink term, W/m³
 w_f = Fracture aperture (width), m
 k_f = Fracture permeability, m²
 p_w = Wellbore pressure, Pa
 ρ_w = Wellbore fluid density, kg/m³
 v_w = Wellbore fluid velocity, m/s
 d_w = Wellbore diameter, m
 f_D = Darcy friction factor, -
 $q_{m \leftrightarrow f}$ = Mass flux between matrix and fracture, kg/(m²·s)
 T_{m-f} = Transmissibility between matrix and fracture, m³·s/kg
 $V_{L\kappa}$ = Langmuir volume constant for component κ , mol/kg
 b_{κ} = Langmuir pressure constant, 1/Pa
 p_{κ} = Partial pressure of component κ , Pa

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